Presentation

Group J

Video Link to presentation:

https://universityofexeteruk-my.sharepoint.com/:v:/g/personal/dg468_exeter_ac_uk/Eb--aIwaFyxMqKz-Scap9RIBonsY8d9e1qACmkeTjcR9qg?e=T6Yzv6

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Overview of Olist Store

Brazilian e-commerce platform that links small and medium-sized companies to online marketplaces

Database system consists of users that are customers as well as sellers

For Customers:

- Customers select and order products from seller-provided list
- Customers provide address info, stored in database
- Unique customer ID generated and stored in database
- Payment options: installments or direct payment
- Payment method info recorded in database
- Customers can track order status via database (shipping and delivery updates)

For Sellers:

- Sellers can list products in the database
- Sellers can view and manage customer orders via the database
- Order status (including delivery info) is recorded in database
- Sellers can track delivery status to ensure timely delivery

Use Case Scenario

- To achieve effective product management, OList store requires a dashboard with useful features
- Provide valuable insights into product performance
- Offer a thorough overview of all items present
- Include information on most popular items sold and average selling price for each goods
- Provide valuable insights into product performance
- Interface must have an easy way to modify merchandise details
- Must be simple to track and handle client feedback on purchased goods

Relational Database - Microsoft SQL Server

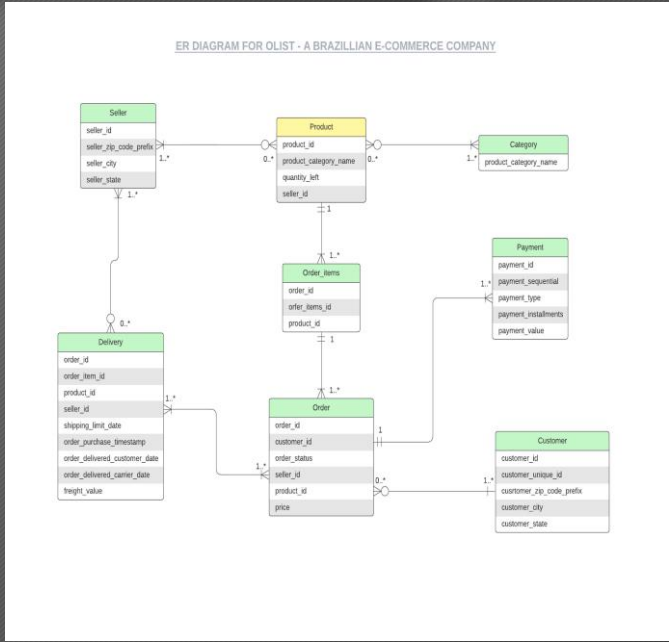
Using an RDBMS, users can create, maintain, and communicate with a relational database, allowing for data tabling

RDBMS supports multi-step procedures facilitates data analysis for Olist

RDBMS follows ACID qualities during transactions despite interruptions

Consistency over availability in RDBMS is crucial for Olist's updated product information for customer orders (CAP Theorem)

RDBMS prevents order misplacement by displaying recent shipping updates to seller



NoSQL Database - Mongo DB

Mongo DB stores data in collections

Ideal for keeping data of various lengths and forms, such as evaluations and product details

Two separate collections can be created for product descriptions and reviews

Data of various lengths and forms, such as product descriptions and evaluations, are ideal for keeping

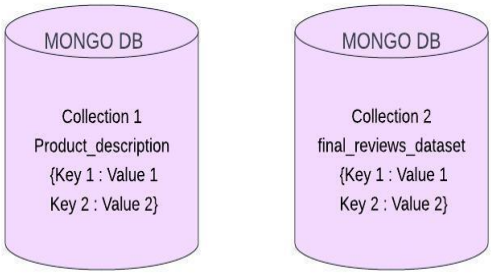
Product descriptions and evaluations can be separated into distinct collections

Product descriptions can be used to store information about each object, such as its name, summary, price, and category

Each product description document in the collection depicts a unique object

Each document in the reviews collection reflects a unique review and includes details such as review content, rating, and the customer who submitted the review

Each review document can include a product ID entry to help connect evaluations to their corresponding products



Polyglot Persistence

Polyglot persistence is a design approach that employs multiple data storage technologies in a single application.

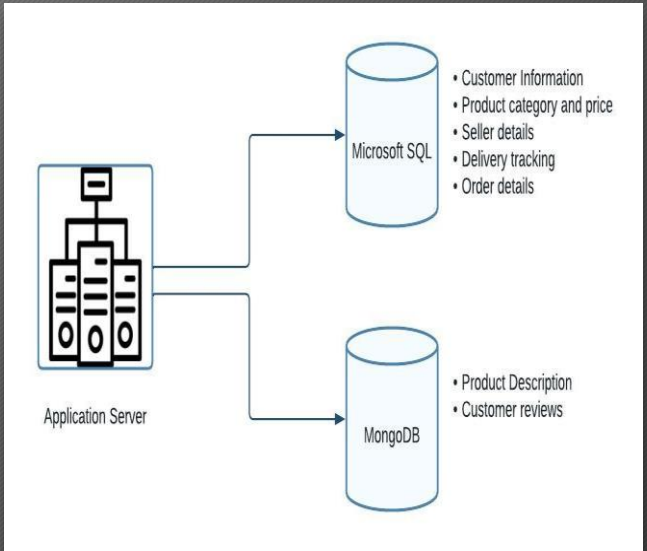
The selection of data storage technologies is based on specific characteristics and intended manipulation of data.

RDBMS and NoSQL databases can complement each other and adopt features from each other when used in demanding applications like Olist.

RDBMS databases are suitable for structured data that fits neatly into tables with predefined relationships, such as customer data, product information, order information, and payment details.

NoSQL databases have a flexible schema, making it easier to accommodate unstructured data and can handle a high volume of writes and reads, such as user reviews.

Polyglot persistence can help Olist build a flexible and scalable application that can handle a wide variety of data storage needs while avoiding the limitations of a single database model.



Conclusion and Recommendations

Recommendation Engine: Customers are recommended goods based on their purchasing experience, increasing sales and satisfaction.

Sentiment Analysis: Analyze customer reviews to better the platform and assist sellers in knowing customer requirements and preferences

View and Click Data Analysis: Utilize data to find high-engagement goods and send tailored online advertisements to promote purchases

Other Platforms Integration: Integrate with apps, search engines, and other platforms to broaden your influence and draw new customers

Own Label Business: Analyze user data and buy trends to develop personalised goods that can outperform market leaders in terms of sales volume

References

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THANK YOU!